

BLY STUDY QUESTIONS

1. Flowerless and seedless plants are also known as _____ [cryptogams]
2. Lower and more primitive plants includes _____ , _____ and _____ [thallophytes, bryophytes and pteridophytes]
3. Adaptation deals with the _____ , _____ or _____ characteristics that enables a living organism to survive and reproduce in a given environment [physiological, structural, behavioral]
4. Hydrophytes are also known as _____ [water loving plants]
5. The aerenchyma in hydrophytes is important for _____ and _____ [gaseous exchange and buoyancy]
6. Mention three examples of hydrophytes and give their respective botanical names
7. Plants adapted to regions with scant or no water and hot conditions are _____ [xerophytes]
8. Xerophytic plants possess _____ to prevent water loss [thick cuticle]
9. Mention four examples of xerophytic plants
10. _____ possess special ecological adaptation to reduced light intensity [sciophytes]
11. Shade-loving plants possess _____ that reflects outgoing light back into the mesophyll [red abaxial layer]
12. Mention three examples of shade-loving plants and give their respective botanical names
13. Sun-loving plants are also known as _____ [Heliophytes]
14. List two characteristics of sun-loving plants
15. Mention two examples of sun-loving plants
16. _____ are plants that can tolerate growing under saline conditions [Halophytes]
17. List three example of sat-loving plants and their respective botanical names
18. Epiphytes are also known as _____ [air plants/tree dwellers]
19. The mesophyll in epiphytic plants is commonly differentiated into _____ and _____ [hypodermis and chlorenchyma]
20. Mention two examples of epiphytes
21. Lithophytes are _____ [plants with special adaptation to growing on rocks or in rocky terrains having scanty humus]
22. List two examples of lithophytes
23. Metallophytes are _____ [plants with special adaptation to natural habitats with toxic level of metals]
24. List three examples of metal-tolerant plants
25. The term higher plant is used for the group of plants that belongs to the division _____ [spermatophyte]
26. Higher plants are characterized by the production of _____ [seeds]
27. _____ is a special structure employed for the reproduction of higher plants [seeds]
28. Higher plants are characterized by the alternation of _____ [generation]
29. Spermatophyta is sub-divided into _____ and _____ [gymnosperm, angiosperm]

30. _____ are less highly developed of the two main sub-divisions of spermatophytes [gymnosperm]
31. There are _____ main classes of gymnosperms [three]
32. The female gametophytes of gymnosperms bear eggs in _____ [archegonia]
33. Gymnosperms have a _____ dominant life cycle [sporophyte]
34. In gymnosperms microspores and megaspores are produced in _____ and _____ respectively [pollen cones, ovulate cones]
35. Angiosperms consists of at least _____ living species classified in _____ families [260,000 and 453]
36. In angiosperms the ovules are enclosed in a _____ [carpel]
37. In angiosperms, double fertilization leads to the formation of _____ [endosperm]
38. Phloem tissue in angiosperms is composed of _____ and _____ [sieve tube, companion cell]
39. The largest group of angiosperms are _____ [dicots]
40. Perennial plants may be woody or herbaceous [true/false]
41. _____ plants do not flower in its first year but creates food storage organs
42. In angiosperms, double fertilization results in _____ and _____ [zygote, endosperm]
43. The division of an organism into identical left and right halves in only one plane is termed _____ [bilateral symmetry]
44. The division of an organism into two identical halves from any plane passing through the central axis of its body is termed _____ [radial symmetry]
45. In an earthworm, the skin or body wall around the 14th to the 16th segment is called _____ [clitellum]
46. The first metamere of an earthworm where the mouth is situated ventrally is called _____ [peristomium]
47. In arthropods, the body is segmented into _____ and _____ [cephalothorax, posterior abdomen]
48. The body of a snail consists of _____, _____, _____ and _____ [head, mantle, foot, visceral mass]
49. In a starfish, the upper surface where the anus is present is called _____ [aboral surface]
50. The body of a fish is divided into _____, _____ and _____ [head, trunk, tail]
51. In a true fish, unpaired fins are _____, _____ and _____ while paired fins are _____ and _____ [(dorsal, ventral, caudal), (pectoral, pelvic)]
52. In a fish, _____ pairs of gill slits are present laterally between _____ and _____ [five, mouth, pectoral fin]
53. In a frog, several rows of spiny teeth are present in the _____ and absent in the _____ [upper jaw, lower jaw]
54. In a frog, the forelimbs have _____ clawless digits and the hindlimbs have _____ clawless digits [four, five]
55. For lizards, the body is divided into _____, _____, _____ and _____ [head, neck, trunk, elongated tail]
56. Lizards possess two pairs of limbs that have _____ clawed digits [five (5)]

57. In Aves, their streamlined body is divided into _____, _____ and _____ [head, neck, trunk]
58. In Aves, the forelimbs are modified into _____ and hindlimbs have _____clawed digits [wings, 4]
59. plants are _____ organisms adapted for life on land [multicellular photosynthetic]
60. In order for plants to be adapted to life on land, they are differentiated into _____ and _____ [hold fast and upper photosynthetic area]
61. Divisions in plant kingdom is _____ in number, it includes _____ bryophytes, _____ seedless plants, _____ gymnosperm and _____ angiosperm [12, 3, 4, 4, 1]
62. The three main groups of flowerless and seedless plants are _____, _____ and _____ [Thallophyta, Bryophyta, Pteridophyta]
63. _____ are the amphibians and pioneers of the plant kingdom [bryophytes]
64. In pteridophytes, _____ contains the reproductive spores [sorus /sori/sporangia]
65. Organisms that live rooted to one spot are _____ while organisms that moves around are _____ [sessile, motile]
66. In a poriferan, the mouth is called _____ [ostia]
67. In sponges, the anus is called _____ [osculum]
68. _____ have only one opening functioning as both mouth and anus [coelenterates or cnidarians]
69. Fish, snake and crocodile can be grouped together for possessing _____ [scales]
70. In body symmetry, segmentation is also called _____ [metamerism]
71. _____ is used for defense and food capture in coelenterates [Cnidoblast]
72. The anterior sucker in a platyhelminth acts as an auctorial organ for _____ and _____ [adhesion, ingestion]
73. Anus is present in a tape worm (true/false) [false]
74. in annelids, _____ is a chitinous structure about the middle of each segment which helps in locomotion and food collection [setae or chaetae]
75. in annelids, the genital or copulative papillae are in the _____ and _____ segments [17th, 18th]
76. in echinoderms, _____ is a sieve-like structure that regulate the amount of water in the water vascular system [madreporite]
77. A marine fish with cartilaginous endoskeleton is called a _____, the exoskeleton has _____ scales [Chondrichthyes, placoid]
78. A marine or fresh water fish with a bony endoskeleton is called an _____, the exoskeleton has _____ or _____ scales [Osteichthyes, cycloid, ctenoid]
79. Amphibians do not possess exoskeleton (true/false) [true]
80. Reptiles are without skin gland (true/false) [true]
81. In Aves, _____ serves as exoskeleton and jaws are modified into _____ [feather, beak]
82. In Aves, the eyes consist of three eyelids, they are _____, _____ and _____ [upper, lower, nictitating membrane]
83. Mammals have _____ as exoskeleton [hair]
84. _____ is where the basic needs of the organism to survive are met [Ecosystem]
85. _____ makes up more than half of all vertebrate species [Pisces]

86. The Pisces belongs to the phylum _____ and sub-phylum _____ [Chordata, vertebrata]
87. Fishes maintain a constant internal body temperature [true/false] [false]
88. A true fish possesses _____ and _____ [backbone and fin]
89. Pectoral and pelvic fins are used for _____, _____ and _____ [steering, balance and braking]
90. _____ fins keep a fish from rolling over in water [dorsal fin]
91. A Fast- swimming fish have _____ tails that provides the thrust needed to speed through water [narrow forked]
92. Slower-swimming fish have a _____ tail that helps them swim around rocks or reefs and catch prey [wide square-shaped]
93. _____ is a flexible bony plate that protects the sensitive gills [operculum]
94. _____ help fishes to move up and down in water [swim bladders]
95. Amphibians have limbs that evolved from _____ and _____ [pectoral and pelvic fins]
96. _____ are the only vertebrates to undergo complete metamorphosis [amphibians]
97. The skin of a frog contains both _____ and _____ [mucus, poisonous gland]
98. A frog has _____ chambered heart with _____ atria and _____ ventricle [3, 2, 1]
99. Mention three adaptive features of an arboreal vertebrate
100. _____ systematics originates from the Greek word _____ which means _____ [systema, placing together]
101. _____ the systematics means _____ of living things in accordance with their _____ [classification, natural relationship]
102. _____ phylum protozoa consist of four classes namely _____, _____, _____ and _____ [Rhizopoda, Ciliata, Flagellata, Sporozoa]
103. _____ is the most important taxon that represents a natural unit [specie]
104. _____ Single celled organisms are called _____ while multicellular celled organisms are called _____ [protozoans, metazoans]
105. _____ Multicellular organisms without tissue grade are called _____ while those with tissue grades are called _____ [parazoans, eumetazoans]
106. _____ Eumetazoa is sub-divided into two groups namely _____ and _____ [diploblastic, triploblastic]
107. _____ Diploblastic animals have _____ and _____ as the two layers in the body wall [endoderm, ectoderm]
108. _____ Triploblastic animals have _____, _____ and _____ as three layers in the body wall [ectoderm, mesoderm, endoderm]
109. _____ For protozoans, nutrition is either _____ or _____ [autotrophic, heterotrophic]
110. _____ The body wall of a poriferan (sponges) contains _____ [spicules]
111. _____ In coelenterates (cnidarians), the endoderm is separated from the ectoderm by _____ [non-cellular mesoglea]
112. _____ Coelenterates are divided into three classes namely _____, _____ and _____ [Hydrozoa, Scyphozoa, Anthozoa]
113. _____ In Platyhelminthes, _____ is responsible for excretion and osmo-regulation [flame cell]

114. Platyhelminthes are divided into three classes namely _____, _____ and _____ [turbellaria, Trematoda, Cestoda]
115. The body cavity of nematodes is considered as a _____ [Pseudo-coelom]
116. In Annelids, excretion and osmo-regulation are achieved by ciliate tubules called _____ [nephridia]
117. _____ are the most successive group of animals [arthropods]
118. The body of an arthropod is covered by hard exoskeleton made of _____ [chitin]
119. The shedding of exoskeleton in arthropods is known as _____ [moulting/ecdyasis]
120. In arthropods, the compound eyes are made up of several photoreceptor sub-units called _____ [ommatidia]
121. In Arthropods, the body cavity is filled with a fluid called _____, such body cavity is known as _____ [Haemolymph, Haemocoel]
122. Mention the classes of arthropods
123. _____ are considered to be the second largest group of animals with regards to specie number [Mollusca]
124. In Molluscs, respiration occurs through _____ [gills/ctenidia]
125. The smallest vertebrate is a fish known as _____ [the Philippine goby]
126. The phylum Chordata is sub-divided into four sub-phyla namely _____, _____, _____ and _____ [hemichordate, urochordate, cephalochordate, vertebrata]